



University of
Nottingham

UK | CHINA | MALAYSIA

COMP4139

Machine Learning (MLE)

Scientific Writing for Machine Learning

Dr Shreyank N Gowda
Dr Direnc Pekaslan

School of Computer Science
University of Nottingham



Top Publication Venues for Machine Learning

- **Top International Conferences:**
 - Neural Information Processing Systems (NIPS)
 - AAAI conference on Artificial Intelligence (AAAI)
 - International Conference on Machine Learning (ICML)
 - The International Conference on Learning Representations (ICLR)
- **Prestigious Journals:**
 - IEEE Transactions on Pattern Analysis and Machine Intelligence
 - IEEE Transactions on Neural Networks and Learning Systems
 - Journal of Machine Learning Research
 - Pattern Recognition
 - Neurocomputing



Paper Format

IEEE Style

Preparation of Papers in Two-Column Format for Conference Proceedings Published by IEEE*

Bart Simpson and Homer Simpson

Department of Nuclear Power Engineering
University of Springfield
Springfield, Nostate 12345, USA

{bart.simpson & homer.simpson}@uspringfield.edu

Monkey King, Bajie Zhu and Seng Tang

Department of Intelligent Robotics
University of Huaguoshan
Huaguoshan, Jilishijie Province, China
monkey.king@shuaguoshan.edu.cn

Abstract - These instructions give you the basic guidelines for preparing papers for IEEE conference proceedings.

Index Terms - List key index terms here. No more than 5.

I. INTRODUCTION

Your goal is to simulate the usual appearance of papers in IEEE conference proceedings. For items not addressed in these instructions, please refer to the last issue of your conference's proceedings for reference or ask your conference Publications Chair for instructions.

A. Preparing Your Paper

1) **Paper Size:** Prepare your paper in full-size format on US letter size paper (8.5 by 11 inches).

2) **Type Sizes and Typefaces:** Follow the font type sizes specified in Table I. The font type sizes are given in points, same as in the MS Word font size points. Times New Roman is the preferred font.

3) **Paper Margins:** Paper margins on the US letter size paper are set as follows: top = 0.75 inches, bottom = 1 inch, side = 0.625 inches. Each column measures 3.5 inches wide, with a 0.25-inch gap between the two columns.

4) **Paper Styles:** Left- and right-justify the columns. On the last page of your paper, adjust the lengths of the columns so that they are equal. Use automatic hyphenation and check spelling and grammar. Use high resolution (300dpi or above) figures, plots, drawings and photos for best printing result.

TABLE I
TYPE SIZE FOR PAPERS

Type size (pts.)	Appearance		
	Regular	Bold	Italic
6	Table superscripts		
8	Section titles ¹ , references, tables, table names ² , table captions, figure captions, footnotes, text subscripts, and superscripts		
9		Abstract, Index Terms	
10	Authors' affiliations, main text, equations, first letter in section titles ³		Subheading
11	Authors' names		
22	Paper title		

¹Uppercase

B. Preparing Your PDF Paper for IEEE Xplore[®]

Detailed instructions on how to prepare PDF files of your papers for IEEE Xplore[®] can be found at

<http://www.ieee.org/pubs/confpubcenter>

PDF job setting files for Acrobat versions 4, 5 and 6 can be found for downloading from the above webpage as well. The instructions for preparing PDF papers for IEEE Xplore[®] must be strictly followed.

II. HELPFUL HINTS

A. Figures and Tables

Try to position figures and tables at the tops and bottoms of columns and avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be centered below the figures; table captions should be centered above. Avoid placing figures and tables before their first mention in the text. Use the abbreviation "Fig. #," even at the beginning of a sentence.

Figure axis labels are often a source of confusion. Use words rather than symbols. For example, as shown in Fig. 1, write "Magnetization," or "Magnetization (M)" not just "M." Put units in parentheses. Do not label axes only with units. In the example, write "Magnetization (A/m)" or "Magnetization (A m⁻¹)." Do not label axes with a ratio of quantities and units. For example, write "Temperature (K)," not "Temperature/K."

Multipliers can be very confusing. Write "Magnetization (kA/m)" or "Magnetization (10³ A/m)." Figure labels should be legible, at 8-point type.

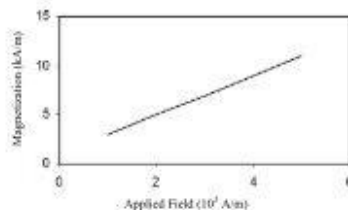


Fig. 1 Magnetization as a function of applied field.
Note how the caption is centered in the column.

Springer Style

Annali dell'Università di Ferrara (2006) 12: 345-678
DOI 10.1007/s11565-002-6789-0

GENERIC ARTICLE

Sample of the standard layout for journal articles

A subtitle is optional. If there is one, write it here

First Author - Second Author - Third Author

Received: 1 December 2003 / Accepted: 12 February 2004 / Published online: 15 March 2004
© Springer-Verlag 2003

Abstract An abstract is required for regular articles. It is optional for other types of articles, e.g., for an editorial. It is always called "Abstract" in English, "Zusammenfassung" in German, and "Résumé" in French. Keywords and Mathematics subject classification and PAC numbers are optional. Which ones are included for a particular article is defined per journal. Lists of nomenclature, symbols, or abbreviations are optional and only included in some journals. Note that although all of these options are possible, it is recommended that each journal doesn't have too many of them.

Keywords First keyword - Second keyword - More

Mathematics Subject Classification (2000) 65K05 · 90C35

1 Introduction

It is common to name the first section of an article "Introduction". This is, however, always optional. Headings of the first three section levels are numbered. This introduction is an example of a first-level heading.

The rest of this section is filler text to make the introduction a little longer. More specifically, we have given a set V of nodes and traffic demands between certain pairs of these nodes. Each demand represents a certain amount of point-to-point traffic to be routed in the network between origin and destination nodes. In addition, a set of

Grants, communicated-by lines, or other notes about the article will be placed here between rules. Such notes are optional.

F. Author - T. Author
Konrad-Zuse-Zentrum für Informationstechnik Berlin, Heilbronner Strasse 10, 10711 Berlin, Germany
E-mail: {f.author,t.author}@zib-berlin.de

S. Author (✉)
Norwegian Telecom Research, P.O. Box 83, 2007 Kjeller, Norway
E-mail: authors@tf.tele.no

* This work is partially supported by NSF Grant #2003168 to H. Simpson and CNSF Grant #9972088 to M. King.



- **Title:**
 - accurate and concise
- **Authors:**
 - order normally reflect contributions
- **Abstract:**
 - motivation; aim; method; contribution; result
- **Introduction**
 - background; aim/objectives; existing solutions; contribution
- **Related work**
 - closely related methods in literature



- **Method**
 - Start with an overview of the method, then many subsections to introduce each element.
- **Evaluation**
 - Dataset; experimental design; parameter settings; evaluation metrics; quantitative and qualitative results
- **Discussion**
 - Pros and cons of the method and why
- **Conclusions**
 - Brief summary of research outcome and future work
- **References**
 - Authors, title, volume, issue, page number, year, (DOI)



Equations, Figures and Tables

- Always re-write the equations by yourself (don't copy as an image)
- Always label all equations, figure, and tables.
- Captions in figures and tables need to be self-contained
- Describe variables, measures in figures and tables
- Explain symbols in equations right after each equation.
- Label all axes in figures and explain all plots in the figures.
- Ensure high resolution figure and all elements are readable in a printed A4 paper.



Scientific Descriptions

- Try to avoid using “I”, “We” etc. Use passive tense.
- Don’t use “But”, “And” to start a sentence, using “Additionally”, “However”.
- Avoid using very long sentences, break down into shorter ones.
- **Don’t use code to describe the method, use maths descriptions.**
- Don’t use references as part of the sentence. E.g. “In [1], a new method is”
- Don’t duplicate, you can always refer to certain statement from other sections.
- Put figures, equations and tables into context. E.g. “As shown in Fig. 1,”



- Citing papers needs to be accurate
- Write one or two sentences to describe the paper you cite
- Try to avoid citing unreliable sources (e.g. websites, blogs), cite peer-reviewed conference and journal papers (avoid arXiv papers as they're not peer-reviewed)
- Don't directly copy figures or sentences from the paper you cite (even you cite them).



Integrity and Plagiarism

- Don't use sentences from other papers. Every conference/journal uses *Turnitin* to screen the paper first.
- Ensure the code and paper are consistent with each other.
- No manipulation of the results, figures and tables.
- Someone can implement the method by reading the paper and get the same result. This is why people also publish their code along with the paper.

Paper Reviewing Process

- Conference
 - Submit by a deadline each year
 - Each paper is reviewed by 2-3 reviewers (mostly double blinded) using a standard form
 - Some conferences have a rebuttal process to defend your paper.
 - Area Chairs will make a final decision to accept or reject the paper based on the comments from reviewers
 - If accepted, travel to a nice place and present/networking
- Journal
 - No deadline, submit anytime of the year.
 - Associate editor will screen it first to check plagiarism, format and research scope.
 - Associate editor will invite a few reviewers to provide comments
 - Outcome: reject; major revision; minor revision; accept.
 - Have the chance to revise the paper, if not rejected straightaway.